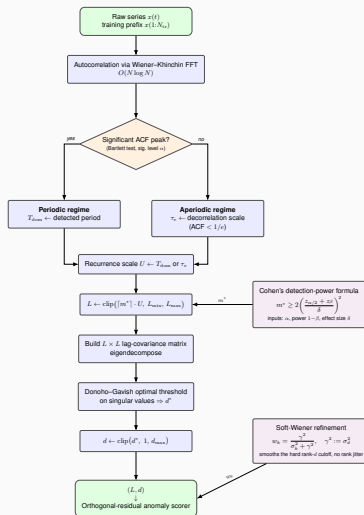


Closed-Form (L, d) Selection

Deriving the delay-embedding window length and subspace rank from first principles

Overview



The Problem

- Series $x(t)$, healthy training prefix $x(1:N_{\text{tr}})$.
- Delay-embed into windows:

$$w_t = [x(t), x(t + \tau), \dots, x(t + (L - 1)\tau)] \in \mathbb{R}^L$$

- Fit a d -**dimensional** healthy subspace $U_d \subset \mathbb{R}^L$ from these windows.
- Before anything else: choose L (window length) and d (subspace rank).

Two numbers. Zero free parameters left unexplained.

Step 1 — Autocorrelation

$$\rho(\ell) = \frac{1}{N \operatorname{Var}(x)} \mathcal{F}^{-1}(|\mathcal{F}(x)|^2)(\ell)$$

- Wiener–Khinchin: power spectrum $\leftrightarrow$ autocorrelation.
- $O(N \log N)$, not the naive $O(N^2)$.

Step 2 — Is There a Real Period?

$$z_{\text{crit}} = \frac{\Phi^{-1}(1 - \alpha/2)}{\sqrt{N}}$$

- Bartlett: under white noise, $\text{SE}(\rho_k) \approx 1/\sqrt{N}$.
- Peak counts only if $\rho(\text{peak}) > z_{\text{crit}}$.

Step 3a — Periodic Regime

$T_{\text{dom}} \leftarrow \text{lag of the strongest significant peak}$

- The series' fundamental period.

Step 3b — Aperiodic Regime

$$\tau_e = \min\{\ell : \rho(\ell) < 1/e\}$$

- Decorrelation scale — the process's memory horizon.

Step 4 — One Recurrence Scale

$$R \leftarrow T_{\text{dom}} \text{ or } \tau_e$$

- Same downstream rule, either regime.

Step 5 — How Many Repeats?

$$m^* \geq 2 \left(\frac{z_{\alpha/2} + z_{\beta}}{\delta} \right)^2$$

- Standard two-sample detection-power formula (Cohen, 1988).
- Inputs: significance α , power $1-\beta$, effect size δ .

δ cannot be estimated from healthy data alone.

Step 6 — Window Length L

$$L = \text{clip}(\lceil m^* \rceil \cdot R, L_{\min}, L_{\max})$$

L = window length, in samples

m^* = required cycle count (Step 5)

R = recurrence scale: T_{dom} or τ_e (Step 4)

$L_{\min} = 2$ (hard floor)

$L_{\max} \approx N_{\text{tr}}/4$ (stability ceiling)

- Integer multiple of the recurrence scale.
- Motivated by SSA separability theory.

Step 7 — Healthy Subspace

$$C = \frac{1}{M} H_{\text{tr}}^{\top} H_{\text{tr}} \in \mathbb{R}^{L \times L}, \quad s_i = \sqrt{M \sigma_i^2}$$

$H_{\text{tr}} \in \mathbb{R}^{M \times L}$ = stacked training windows

M = number of training windows (rows of H_{tr})

σ_i^2 = eigenvalues of C (descending)

s_i = singular values of H_{tr}

- Lag-covariance eigendecomposition of the training windows.
- Signal concentrates in a few directions; noise spreads evenly.

Step 8 — Signal vs. Noise, Automatically

$$\tau_* = \omega(\beta) \cdot \text{median}(s), \quad \beta = L/M$$

$$\omega(\beta) \approx 0.56\beta^3 - 0.95\beta^2 + 1.82\beta + 1.43$$

$$d^* = \#\{i : s_i > \tau_*\}$$

- Donoho & Gavish (2014) optimal hard threshold.
- No prior knowledge of the noise level required.

Step 9 — Capping the Rank

$$d = \text{clip}(d^*, 1, d_{\max})$$

- Uncapped: healthy subspace can absorb part of an anomaly too.
- “Expressive slack” — observed, not just theorized.
- $d_{\max}$: tuned once on labelled data, not derived.

$$r_{\perp} = (I_L - U_d U_d^{\top}) w, \quad \|w\|_2^2 = \|U_d^{\top} w\|_2^2 + \|r_{\perp}\|_2^2$$

$$\text{soft-Wiener: } P_{\perp}^{(\gamma)} = \sum_k w_k u_k u_k^{\top}, \quad w_k = \frac{\gamma^2}{\sigma_k^2 + \gamma^2}$$

$$r_{\perp}^{(\gamma)} = P_{\perp}^{(\gamma)} w \quad (\text{same } w \text{ — only the projector softens})$$

- U_d : healthy subspace (top- d eigenvectors, Step 7).
- Anomaly score = $\|r_{\perp}^{(\gamma)}\|_2$ — energy *outside* the subspace.

Summary

- $T_{\text{dom}}, \tau_e, d^*$ — derived, unsupervised.
- δ, d_{max} — the only two constants needing labelled data.

Closed-form. Zero trained parameters.